

AS.410.800.01
Independent Research in Biotechnology

A Platelet Mimetic-Based Approach to Detect Markers of Restenosis

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Degree Program
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A. ABSTRACT

The effective prevention of restenosis among patients likely requires an understanding of the molecular mechanisms governing the process, which are unclear. There is evidence, however, that a link to clotting exists. In mouse models of restenosis, two important regulators of clotting—serotonin and tissue factor (TF)—have been implicated in effects on blood vessel smooth muscle cells, contributing to vascular matrix degradation and vessel wall closure.

This research plan aims to develop novel nanosensors to enable continuous monitoring of serotonin and TF in a perfusion chamber system containing whole blood from mouse models of restenosis. The nanosensors will be specifically targeted to blood platelets comprising the growing occlusive clot, where serotonin and TF exert a maximal effect on the vascular matrix. The nanosensors will also mimic the size and mechanical properties of platelets, allowing them to effectively incorporate into the clot for optimal imaging.

The specific aims of this research are:

- 1) *Develop and characterize dual optical nanosensors for serotonin and TF detection.*
- 2) *Develop and characterize platelet mimetic nanosensors for serotonin and TF.*
- 3) *Test nanosensor ability to detect serotonin and TF in a flow chamber system with mouse whole blood.*

The spatial and temporal resolution provided by this approach will enable the direct correlation of serotonin release and TF-mediated initiation of clotting to the development of acute clotting malfunctions in restenosis. This quantitative association of serotonin concentrations to TF-mediated clotting activity will serve as a predictive measure for the propensity to develop restenosis. The information obtained from the proposed study will not only provide a useful tool to advance our understanding of the mechanisms of restenosis but will also provide a basis for the development of nanoscale diagnostic reagents.

B. SPECIFIC AIMS

The effective prevention of restenosis requires an understanding of the molecular mechanisms governing the process. These mechanisms are largely unclear, but there have been key links identified between clotting and restenosis that could hold the key to a solution. Of the major modulators of clotting, serotonin and tissue factor (TF) released from activated blood platelets in a damaged area or on a disrupted plaque could play a critical role in the development of occlusive clots, however the individual and collective contributions of TF and serotonin to the development of restenosis are unclear.

This study will develop and characterize platelet-targeted nanosensors that can quantitatively detect changes in serotonin concentrations released from platelets using enzyme-based sensors, while simultaneously determining platelet-stored TF expression and TF activity using a fluorescent peptide-based approach. These nanosensors will be formulated to mimic the mechanical properties, size, and surface molecule composition of blood platelets, which will enable them to be targeted to growing clots and in the presence of large amounts of newly-released serotonin and TF. The developed nanosensors, employed in a flow chamber diagnostic tool, may be used to elucidate the role of serotonin and TF in a mouse model of restenosis by providing quantitative information about the two molecules relative to disease progression.

Specific Aim 1. Develop and characterize dual optical nanosensors for serotonin and TF detection.

We will fabricate the nanosensors, tune their responsive range, and characterize their response to serotonin and TF.

Metrics: The responsive range of the nanosensors will be tuned to the physiological ranges of serotonin and TF, and reversibility and specificity over competing analysis will be assessed in physiologically-relevant concentration ranges.

Specific Aim 2. Develop and characterize platelet mimetic nanosensors for serotonin and TF.

Using the nanosensor formulation from **Aim 1**, platelet mimetic nanosensors will be engineered to exhibit adhesive properties for selective recognition of platelet-rich clots. Nanosensors will be coated with molecules that bind platelet receptors and modified to mimic the ellipsoid shape and flexibility of platelets.

Metrics: The efficiency of the interaction of platelet mimetic particles with platelets will be assessed quantitatively by measuring the duration of particle-surface contacts and deriving the corresponding apparent dissociation rate constant (k_{off}). The serotonin and TF released from the platelets will also be determined.

Specific Aim 3. Test nanosensor ability to detect serotonin and TF in a flow chamber system with mouse whole blood.

Testing sensor function in a research model is critical to development of a useful sensor platform. This aim will take the platelet mimetic nanosensors developed in **Aim 2** and demonstrate the ability to continuously monitor serotonin and TF concentrations in blood drawn from normal mice and mouse models of restenosis. In wild-type mice, blood will be collected and changes in externally-added serotonin and TF will be assessed. Mice will be administered high fat diets, and atherosclerotic mice will be treated by balloon angioplasty.

Metrics: At various durations of time following balloon angioplasty, the rate and total accumulation of serotonin and TF will be recorded. The ratio of the quantitative measure of serotonin concentration vs. TF/ fVIIa activity will be calculated and assessed for a correlation to progression of restenosis, as determined by coronary artery section staining.

The strategy outlined in **Aim 1** is designed to generate nanosensors that can detect serotonin and TF in relevant physiological concentrations. The experiments outlined in **Aim 2** are expected to provide novel platelet-targeted nanosensors that are derived to dually detect serotonin and TF *in vitro*. The experiments outlined in **Aim 3** are expected to provide a rational basis for using the developed nanosensors as a potential diagnostic tool by establishing a flow chamber-based method to detect serotonin and TF in mouse whole blood.

C. BACKGROUND & SIGNIFICANCE

Atherosclerosis, which is the dominant cause of cardiovascular disease (CVD), can be characterized as an inflammatory response to endothelial injury (1-2). This disease, which is responsible for more deaths in the United States than any other condition (3), is characterized by thickening of the arterial walls causing an impendence of blood flow (4). Percutaneous coronary interventions (PCIs)—such as balloon angioplasty (5) and coronary stents (6)—have revolutionized the effective management of atherosclerosis and its symptoms (4). However, the superimposition of mechanical injury on coronary atherosclerosis initiates a new disease process that may complicate the patient's clinical course. This new disease process, termed restenosis, is the major complication that occurs after PCIs (7-8) and results in re-narrowing of the blood vessel (9). The effective prevention of restenosis among patients likely requires an understanding of the molecular mechanisms governing the process (10-12). These mechanisms are largely unclear, however the potential roles of thrombosis in atherosclerosis provides key clues.

This research plan describes the development of novel nanosensors to enable continuous monitoring of key molecules involved in thrombosis. The use of these nanosensors in a mouse model of restenosis will enable us to decipher the key mechanisms in thrombosis that affect the development of restenosis. By focusing on two key modulators of thrombosis—serotonin and tissue factor (TF)—that have been implicated in disease progression in atherosclerosis and restenosis, we aim to achieve the spatial and temporal resolution necessary to enable the direct correlation of serotonin release and TF-mediated initiation of coagulation to the development of acute thrombotic events in cardiovascular disease. This quantitative association of serotonin concentrations to TF-mediated coagulation activity will serve as a predictive measure for the propensity to develop restenosis following PCIs (Figure 1). The information obtained from the proposed study will not only provide a useful tool to advance our understanding of the mechanisms of clotting disorders, but will also provide a basis for the development of nanoscale testing reagents.

Key modulators of the procoagulant response: Platelets are the main modulators of the procoagulant response and are known to contribute to the development of ischemic complications in cardiovascular disease (13). They recognize and bind to proteins present in the sub-endothelial space to form a platelet aggregate on a damaged surface. Activated platelets within clots produce vascular smooth muscle cell (VSMC) mitogens and thrombin, which perpetuate and amplify smooth muscle cell migration and proliferation.

Platelets are activated by several agonists that bind to specific platelet membrane receptors. Serotonin (5-HT) is an indolamine with vasoconstrictive properties. Platelets possess receptors for serotonin (5-HT_{2A}) (14) in their surface and a serotonin transporter (SERT) to re-uptake serotonin rapidly, move it into platelet dense bodies (15), and secrete it during platelet activation (**Figure 2**). Serotonin is generally present in platelets at concentrations between 0.2 and 5 mM, as measured by western blot (14-15).

Another major modulator of the procoagulant response is tissue factor (TF). TF exposed at sites of vascular damage initiates the coagulation mechanism to lead thrombin generation that activates other platelets and generates a stable fibrin net(16). TF binds to activated coagulation factor VII (fVIIa) resulting in activation of factor IX (fIX) and factor X (fX), ultimately leading to fibrin formation (**Figure 3**). The concentration of soluble TF in blood is thought to be less than 500 pM (16), however the estimated concentration of released TF on the surface of cells is between 1 μ M and 0.5 mM, as measured by immunofluorescence studies (17).

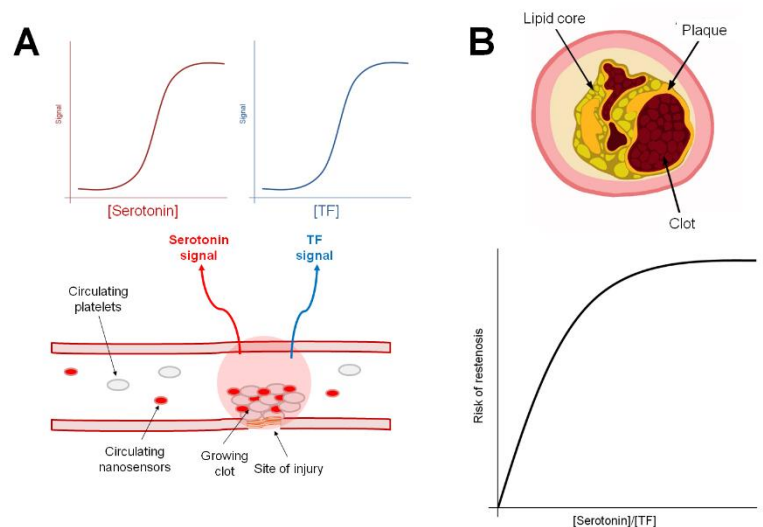


Figure 1: Research strategy for detection of serotonin and tissue factor and association to restenosis.

(A) Nanosensors will be targeted to activated platelets in a growing clot. The serotonin concentration and TF-mediated coagulation activity will be determined by comparing the signal intensity to calibration curves for each analyte. (B) The degree of restenosis in mice will be assessed and correlated to the ratio of serotonin concentration to TF-mediated coagulation activity.

The roles of serotonin and tissue factor in the development of cardiovascular disease:

Several studies have suggested a connection between serotonergic mechanisms and cardiovascular events. Some studies have reported different polymorphisms in the gene for the 5-HT_{2A} receptor (18), and the SERT receptor (19), that are associated with an increased cardiovascular risk (20). Recently, Nishihira et. al. (21) have reported that intravenous injection of sarpogrelate, an inhibitor of the 5-HT_{2A} receptor, in a rabbit model significantly reduced *ex vivo* platelet aggregation and significantly prevented occlusive thrombus formation *in vivo*. Other authors have also reported that increased platelet serotonin could be effective in the control of bleeding in idiopathic thrombocytopenic purpura (22). There is also clinical evidence that depression is commonly present in patients with coronary heart disease (23-24) and, in fact, it is considered an independent factor of cardiovascular risk (25).

Recent studies have shown that platelets possess mechanisms to internalize TF-rich microvesicles (26). These microvesicles could enter through membranes and occasionally reach α -granules, where they could be stored until platelet activation. In mouse models of restenosis, activation of platelets has been shown to result in the exposure of the stored TF, as well as the release of serotonin from platelet granules and dense bodies, respectively. TF and serotonin released from activated platelets in a damaged area or on a disrupted plaque could play a critical role in the development of acute thrombotic events; however the individual and collective contributions of TF and serotonin to the development of restenosis following PCIs are unclear.

D. INNOVATION

A potential role for nanosensor development in thrombosis imaging: *In vivo* thrombosis imaging would benefit from a method that can visualize multiple viewing fields, provide a robust quantitative measure of specific coagulation events, provide spatial and temporal resolution in measuring coagulation events, and continuously monitor thrombotic events in the absence or presence of external stimuli.

The Clark laboratory has a broad interest in *in vivo* detection and tracking of analytes using reversible optical nanosensors. These spherical nanosensors are made of a plasticized hydrophobic polymer core with a hydrophilic coating (27). The laboratory has recently developed nanosensors utilizing an enzyme-based system that can be used to detect histamine (28). This approach was based on significant experience in the design and application of this class of ion sensors predominantly used for electrolyte sensing (29).

Taking advantage of this type of system, we can design nanosensors that have highly desirable characteristics for *in vivo* thrombosis imaging. First, the system is highly tunable because changing the types, concentrations or ratios of internal chemistries can change the response of the sensor. Second, sensor lifetime ranges from days to several weeks. Third, sensor components are modular, and for the detection of a new target recognition elements can be replaced without significant modification of the remaining components of the sensor system. The combination of these advantages provides a significant improvement over standard molecular probes for thrombosis imaging, which include fluorescence microscopy using antibody-labeled proteins or calcium ionophores.

Enzyme optical nanosensors: Research during the past several decades has greatly improved *in vitro* and *in vivo* analysis through the development of enzymatic biosensors (30-31). The range of *in vivo* sensors available for either basic research or clinical diagnostics has continually grown, although the best developed are those for

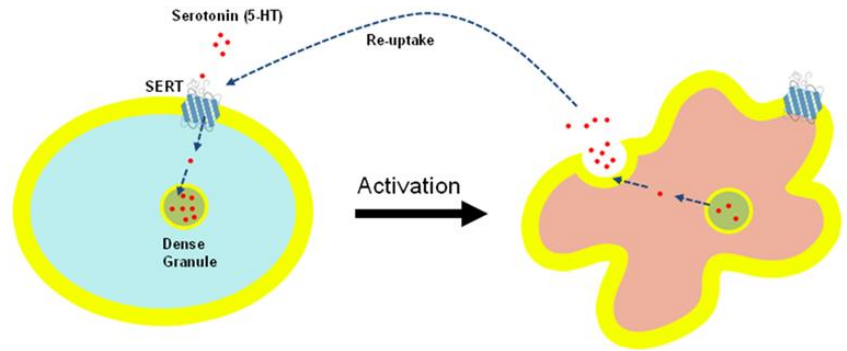


Figure 2: Serotonin storage, release, and re-uptake in platelets.

Serotonin is stored in the dense granules of resting platelets. It is released upon platelet activation and re-uptake is through SERT on the platelet surface.

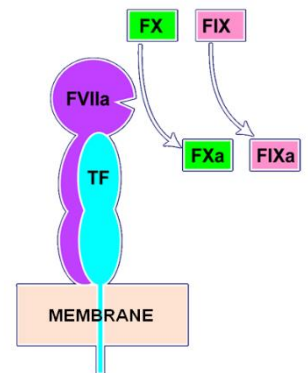


Figure 3: Tissue factor-mediated activation of fX and fIX.

Membrane-expressed tissue factor (TF) binds to activated coagulation factor VII (fVIIa), forming the TF/fVIIa complex that activates coagulation factor IX (fIX) and coagulation factor X (fX).

glucose. More recently, nanosensors have been fabricated based on enzymatic recognition for the detection of histamine (28). These enzyme-based sensors provide the advantages of enzymes while still retaining the benefits of optode-based nanosensors. Coupling these two techniques together enable the detection of a wider range of biological analytes and transduction of this signal into a fluorescent readout through the use of a polymeric nanosensor.

Protease-sensitive optical sensors: The construction of synthetic optical sensors for thrombosis has primarily involved modifying the surface of iron oxide nanoworms (32-33). Specifically, an assay has been developed to detect the activity of the blood serine protease thrombin with cleavable substrate-reporter tandem peptides (34). To develop a suitable substrate, fluorophore-labeled thrombin-sensitive peptides were conjugated onto nanoworms. Freely-emitting peptide fragments that were released by thrombin activity rapidly increased sample fluorescence (34).

E. PRELIMINARY STUDIES

To facilitate growth of platelet progenitor megakaryocytes and endothelial cells, the nanosensors were initially developed as nanofibers. These nanofibers, which are composed of an electrospun polymer solution containing sensor elements, can promote efficient megakaryocyte and endothelial cell growth and release of stored serotonin and TF while providing a platform to detect released serotonin and TF. The preliminary results described below use these nanofibers as the platform. For the proposal, the nanofibers will be reformulated as spherical nanosensors, which will then be manipulated to mimic platelet shape and morphology.

Development and characterization of serotonin-sensing nanofibers: The liver metabolizes serotonin, first by monoamine oxidase-A (MAO-A) oxidation to the aldehyde product (5-HIAL) followed by aldehyde dehydrogenase (ADH) oxidation to the indole acetic acid derivative, 5-HIAA (35)(Figure 4A). Taking advantage of the chemistry of MAO conversion of serotonin, nanofibers were designed to detect serotonin *in vitro*. This monitoring mechanism uses an enzymatic recognition element linked to phosphorescent nanofibers. The oxygen responsive element in these nanofibers is Pt(II) meso-tetra(pentafluorophenyl)porphine (36-38) (PtTFPP), a hydrophobic platinum porphyrin dye, which has a reversible, oxygen-dependent phosphorescent signal. Additionally, the fiber core

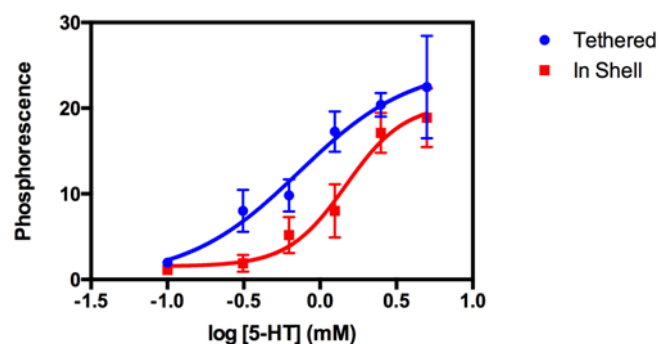


Figure 5: Phosphorescence calibration of nanofibers in response to serotonin.

The calibration curves for nanofibers coaxially electrospun to form a plasticized polymer core and an aqueous shell containing MAO (red) or nanofibers coaxially electrospun to form a plasticized polymer core and an aqueous shell containing collagen, which was covalently linked to MAO (blue) are shown.

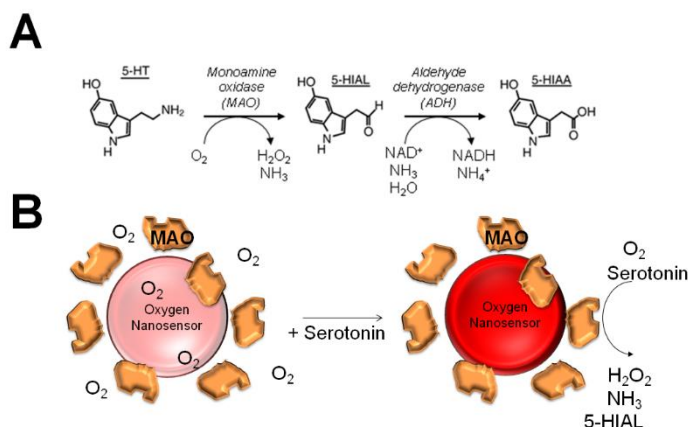


Figure 4: Proposed mechanism of serotonin-sensitive nanosensors.

(A) Serotonin (5-HT) is converted by monoamine oxidase (MAO) and aldehyde dehydrogenase (ADH). **(B)** Serotonin-sensitive nanosensor design.

contains a reference fluorophore, Rhodamine B, which is not sensitive to oxygen concentrations. This enables ratiometric measurements of serotonin, whereby the ratio of the signal from the two dyes will change with serotonin concentrations independent of sensor concentration. MAO consumes oxygen when it converts serotonin into 5-HIAL, H₂O₂, and NH₃. When the enzymatic activity of MAO causes oxygen levels to drop, the oxygen-responsive nanofibers increase in phosphorescence (Figure 4B).

To make the nanofibers sensitive to serotonin, two approaches were taken: 1) nanofibers were coaxially electrospun to form a plasticized polymer core and an aqueous shell containing MAO, and 2) nanofibers were coaxially electrospun to form a plasticized polymer core and an aqueous shell containing collagen, which was covalently linked to MAO. In both approaches, in the absence of serotonin, nanofibers emitted a low phosphorescent signal, which is consistent with oxygen-

induced quenching (**Figure 5**). Upon addition of serotonin, MAO consumes oxygen at different rates ($t_{95\%} = 2.2$ min), which results in phosphorescence emission from the nanofibers.

Development and characterization of thrombin activity-sensing nanofibers: Using two Cy3-conjugated peptides—one specific to thrombin (Suc-O-DPRSFL-c(Cy3)) and another control polyethylene glycol (PEG)-based construct (Suc-(mPEG)-c(Cy3))—nanofibers were designed to detect thrombin activity *in vitro*. These peptides are based on a fluorescence quenching mechanism, in which a succinimidyl moiety serves as the quencher for Cy3 fluorescence. Hydrolysis of the peptide bond between the fluorophore/quencher pair generates fluorescence that permits the measurement of thrombin activity.

In the absence of thrombin, Suc-O-DPRSFL-c(Cy3)-linked nanofibers emitted a low fluorescent signal, which is consistent with low enzyme activity (**Figure 6, circle trace**). Upon addition of thrombin, the peptide is cleaved, which resulted in fluorescence emission from the nanofibers. The control peptide (Suc-(mPEG)-c(Cy3)) was not cleaved in the presence of thrombin (**Figure 6, triangle trace**).

F. RESEARCH DESIGN & METHODS

General strategy: The overall approach employed in this study is to develop and characterize platelet-targeted nanosensors that can quantitatively detect changes in serotonin concentrations released from platelets using enzyme-based sensors, while simultaneously determining platelet-stored TF expression and TF activity using a fluorescent peptide-based approach. These nanosensors will be formulated to mimic the mechanical properties, size, and surface molecule composition of blood platelets, which will enable them to be targeted to growing clots and in the presence of large amounts of newly-released serotonin and TF. The developed nanosensors may be used to elucidate the role of serotonin and TF in a mouse model of restenosis by providing quantitative information about the two molecules relative to disease progression.

Research aims: The strategy outlined in **Aim 1** is designed to generate nanosensors that can detect serotonin and TF in relevant physiological concentrations.

The experiments outlined in **Aim 2** are expected to provide novel platelet-targeted nanosensors that are derived to dually detect serotonin and TF *in vitro*. The experiments outlined in **Aim 3** are expected to provide a rational basis for using the developed nanosensors as a potential diagnostic tool by a flow chamber-based method to detect serotonin and TF in mouse whole blood.

Aim 1. Develop optical nanosensors for serotonin and TF detection and perform *in vitro* characterization of the nanosensors.

This aim will serve to fabricate the nanosensors based on the strategies used in the preliminary data. We will tune their responsive range to 0.1 to 5 mM for serotonin and 50 μ M and 500 μ M for TF. We will also characterize their response to purified serotonin or TF *in vitro*.

Aim 1A. Develop phosphorescence-based sensors for serotonin detection and characterize *in vitro*. Calibration of the nanosensors will be performed similar to the previous results section above. The two intensities at a given concentration will be normalized and plotted against the log of the serotonin concentration and a fit to a sigmoidal curve fit, from which the K_d will be calculated. Sensors will also be tested with varying concentrations of potential contaminants as well as testing in complex samples such as blood serum in order to quantify specificity. The response time of the nanosensors will be assessed using a fluorescence plate reader and measuring the time trace of the fluorescence intensity upon rapid addition of serotonin.

Aim 1B. Develop fluorescence-based sensors for TF detection and characterize *in vitro*. Instead of using Cy3, NIR dyes will be used to synthesize N-terminally substituted NIR peptides. The K_M and k_{cat} values of the conversion of these substrates by TF, fVIIa, and TF/fVIIa will be quantified. Suitable substrates will be used to fabricate NIR peptide-coupled nanosensors specific to TF/fVIIa, which will then be characterized *in vitro* by a fluorescent plate reader. These nanosensors will be calibrated to determine a specific activity of TF/fVIIa based on the K_M and k_{cat} values. Sensors will also be tested with varying concentrations of potential

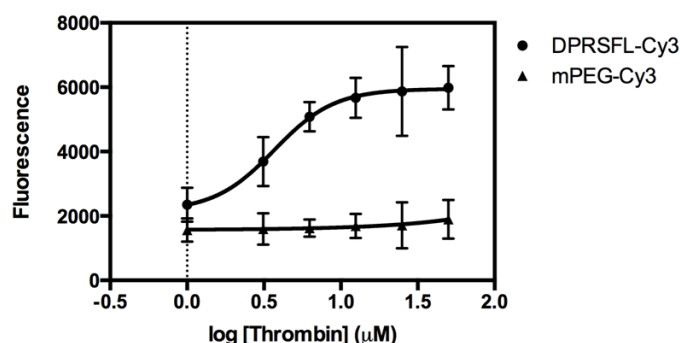


Figure 6: Fluorescence calibration of nanofibers in response to thrombin.

The calibration curves for experimental Suc-O-DPRSFL-c(Cy3)-linked nanofibers (DPRSFL-Cy3) and negative control Suc-(mPEG)-c(Cy3)-linked nanofibers (mPEG-Cy3).

contaminants (other coagulation enzymes) as well as testing in complex samples such as blood serum in order to quantify specificity.

Aim 2. Develop and characterize a platelet-targeted dual-sensing system to quantitatively detect changes in serotonin and TF.

Based on the preliminary data, the fluorescent nanosensors will need reformulation to specifically target the nanosensors to platelets. In addition, modifications to the nanosensor formulation will need to be made to allow dual detection of serotonin and TF by shifting the dynamic range in order to center the linear range of the calibration curves over the physiological analyte ranges.

Aim 2A. Develop platelet-targeted nanosensors and determine their specificity for platelets.

Reports have been published on synthetic blood cells (39-40) and RGD-coated nanoparticles reproducing some of the functions exhibited by platelets (41). Platelets express an array of surface proteins that act as adhesive substrate receptors and are essential for localized hemostatic plug formation. One of these receptors, glycoprotein (GP) Iba—a component of the GPIb-IX-V complex—binds to the A1 domain of multimeric von Willebrand factor (VWF), endowing platelets with the ability to tether to damaged vascular surfaces under shear stress. Platelet-targeting nanosensors will be engineered to exhibit adhesive properties for selective recognition of platelet thrombi by exploiting characteristics of the GPIba/VWF-A1 bond. Initial experiments will be performed with spherical polymeric particles coated through physical adsorption with recombinant, disulfide-linked dimers of VWF-A1 or chimeric dimers of the GPIba amino terminal domain (GPIbaN) containing the VWF-A1 binding site. The coated particles will be tested by perfusing each type over the respective counter-ligand immobilized onto the bottom surface of rectangular flow chambers. The efficiency of the interaction will be assessed quantitatively by measuring the duration of particle-surface contacts and deriving the corresponding apparent dissociation rate constant (k_{off}). Spherical nanoparticles will be stretched on a two dimensional stretcher and the effect of particle shape on the k_{off} of the interaction as a function of shear rate will be evaluated.

Aim 2B. Develop dual-sensing nanosensors for serotonin and TF and characterize their response in vivo. Platelet-targeted nanosensors from the above sub-aim will be loaded with oxygen-sensitive and reference dyes, and the surface of the nanosensors will be modified to be linked to MAO and TF/fVIIa-sensitive peptides. Calibration of the nanosensors will be performed similar to the previous results section above. The intensities at a given concentration of serotonin or TF will be normalized and plotted against the log of the analyte concentration and a fit to a sigmoidal curve fit, from which the K_d will be calculated. Sensors will then be tested for changing concentrations of one analyte in the presence of a constant concentration of the other. Finally, sensors will be tested for serotonin reversibility in the presence of TF/fVIIa.

Aim 3. Test nanosensor ability to detect serotonin and TF in a flow chamber system with purified protein and murine whole blood.

Testing sensor function in a research model is critical to development of a useful sensor platform. This aim demonstrates the ability to continuously monitor analyte concentrations while manipulating those levels experimentally. The BioFlux microfluidic system (42-46) has been widely used to study platelet aggregation under fluid shear stress. This system provides the ability to emulate physiological flow and shear force in an *in vitro* model while maintaining a convenient well plate format.

Aim 3A. Test the serotonin and TF response of nanosensors in a flow chamber system with wild-type mouse whole blood. The microfluidic channels of the BioFlux system will be coated with soluble type 1 collagen and nanosensors of varying concentrations will be added to anti-coagulated whole mouse blood from wild-type C57BL/6 mice and flowed through the collagen-coated surface under fluid shear stress (~ 10 dynes/cm²). Defined concentrations of serotonin and TF will be flowed into the chamber and the rate and total accumulation of serotonin and TF will be recorded every 10 seconds for 5 minutes under a confocal microscope equipped with a high speed digital camera and the data will be analyzed using the BioFlux Montage software (43-44). The intensities at a given concentration of serotonin or TF will be normalized and plotted against the log of the analyte concentration and a fit to a sigmoidal curve fit, from which the K_d will be calculated.

Aim 3B. Test the serotonin and TF response of nanosensors in a flow chamber system with mouse whole blood from a restenosis model. To develop a mouse model of restenosis, C57BL/6 mice will be administered a high fat diet over the course of 1, 6, and 12 hours and 1, 2, and 4 days. Following the duration of high fat diet administration mice will be examined for atherosclerotic lesion development and coronary balloon angioplasty will be performed. At various time points following angioplasty, whole blood will be drawn from the mice. Nanosensors will be added to the anti-coagulated whole mouse blood and flowed through the collagen-coated surface under fluid shear stress (~ 10 dynes/cm²). The rate and total accumulation of serotonin and TF will be recorded and the ratio of the quantitative measure of serotonin concentration vs. TF/fVIIa activity will be

calculated and assessed for a correlation to high fat diet administration or lesion development by Mac-3 antibody staining of excised mouse carotid arteries.

Expected outcomes, anticipated challenges, and alternative approaches: We expect these experiments to demonstrate a quantitative correlation between serotonin and TF dynamics and risk of restenosis in an animal model. Using platelet-mimetic nanosensors that are specifically targeted to growing clots and that have the spatial and temporal resolution to detect physiologically-relevant changes in serotonin and TF, we expect to provide key findings that link thrombosis and restenosis. These nanosensors can be reformulated to assess changes in a multitude of putative modulators of cardiovascular events.

One potential challenge is that the response time may not be fast enough to capture serotonin and tissue factor release dynamics. We have shown that the response time of our nanosensor technology is dependent on size and even at larger diameters (120 nm) is fast enough to monitor single ion channel dynamics in cardiac cells. We anticipate that the nanosensors described in this proposal will be able to respond to dynamics of serotonin and tissue factor release, which are slower than single ion channel dynamics. In addition, serotonin re-uptake by platelets is a slower process than release from dense granules, thus monitoring fluctuations in serotonin due to re-uptake will be possible. If the response time is demonstrated to be too slow, we may formulate new nanosensors with DNA origami-based architectures as opposed to using polymeric nanoparticles. The control of size and density of sensing elements is more tunable with these nanosensor formulations.

Another challenge is that our proposed optical dye may not be usable for imaging in whole blood. This preliminary research utilizes oxygen-sensitive dyes with UV excitation for serotonin sensing, which may cause imaging problems in whole blood due to the short wavelengths and high background. In this instance, Oxphor G2—a porphyrin based near-infrared (NIR, 630 nm excitation) oxygen sensitive dye—can be used. This dye responds to oxygen in sensor systems and also offers the advantage of NIR imaging for better penetration.

Finally, translation of this to in vivo work presents a significant challenge. Fluorescence and phosphorescence imaging would be useful for imaging in the exposed vessels of mice, however imaging these probes in mice without surgery will be difficult because of imaging depth limitations. Although this is outside the scope of the current application, we are collaborating with Lihong Wang at Washington University St. Louis to pair our nanosensors with his photoacoustic imaging system. This will enable measurements that provide quantitative chemical information deeper into tissue than fluorescence measurements can provide.

G. RESEARCH TIMELINE

	M1	M2	M3	M4	M5	M6
Aim 1: Develop serotonin and TF nanosensors	X	X				
Develop and characterize serotonin nanosensors	X					
Develop and characterize TF nanosensors		X				
Aim 2: Develop platelet-targeted dual nanosensors		X	X	X		
Platelet targeting and off-rate analysis		X	X			
Development and characterization of serotonin/TF dual nanosensors			X	X		
Aim 3: Test nanosensors in flow chamber system					X	X
Flow chamber analysis of nanosensors with whole blood					X	
Flow chamber analysis with whole blood from restenosis model						X

H. BUDGET

Category	Items	Description	Total Cost
Materials and Supplies			
Reagents/Supplies	Cells	LK-4 - CRL-2345, hybridoma cell line LK-4	\$850.00
Reagents/Supplies	Blood and blood supplies	Mouse whole blood, platelets, restenosis model	\$1,100.00
Reagents/Supplies	Cell culture media	Gases, buffers, media, and medium supplements for platelet cell cultures	\$900.00
Reagents/Supplies	Nanoparticle formulation solvents	Average cost per test based on 200 assays	\$5,000.00
Reagents/Supplies	Quantum dots	HyperPlus 1/4 volume High Throughput	\$2,050.00
Reagents/Supplies	Surfactants and other fabrication reagents	Nanoparticle formulation surfactants, including PDMS	\$1,100.00
Consumables	Pipette tips, buffers, reagents	Plastic and glass consumables	\$4,000.00
Total Materials and Supplies			\$15,000.00
Lab/Instrument Fees			
Facility Fees	MSE Fabrication facility costs	BioRad CFX96 and 384 Touch, full service	\$4,500.00
Facility Fees	Microscopy facility costs	SEM, TEM, AFM (self-service)	\$4,000.00
Facility Fees	Mass spec facility costs	Agilent BioAnalyzer (\$70 per chip, up to 12 samples per chip)	\$1,500.00
Total Lab/Instrument Fees			\$10,000.00
TOTAL BUDGET			\$25,000.00

I. BUDGET JUSTIFICATION

Materials and Supplies

Materials required to complete the project include: PDMS, surfactant, chemical solvents, cells, culture medium, quantum dots which are used in the cytotoxicity experiments. (\$6,000). Some lab supplies include glassware, tubing, syringes, wipes, pipette tips. (\$4,000). Chemicals and general supplies used for nanoparticle synthesis (\$5,000).

Total Materials and supplies cost: \$15,000.

Lab/Instrument Fees

The following usage fees and training apply at the facilities of Material Science and Engineering (\$4,500), Central facility for Microscopy & Microanalysis (\$4,000), and Mass Spectrometry Lab (\$1,500).

Total Lab/Instrument Fees Cost: \$10,000.

J. REFERENCES

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